HAEMOPHILIA INTELLIGENCE RADAR

MetaRadar From Scattered Signals to Strategic Intelligence

Novo Nordisk GBS Hackathon 2026 | Problem Statement #3 | Rare Disease Pilot

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1. Problem, relevance, intended users, and differentiation

The haemophilia landscape generates signals across clinical trials, scientific publications, pharmaceutical news, regulatory updates, safety communications, congresses, and market-access developments. These sources are fragmented, making it difficult for teams to identify what materially changed and what deserves immediate attention.

Relevance to Novo Nordisk: As a global leader in haemophilia care with key therapies (such as Esperoct, Alprolix, NovoSeven, and investigational denecimig / Mim8) operating alongside evolving non-factor treatments and gene therapies, Novo Nordisk requires rapid, cross-functional intelligence synthesis to maintain clinical leadership, anticipate market shifts, and streamline internal risk review.

MetaRadar addresses this by converting scattered external information into **source-linked, prioritized, and actionable internal intelligence**. It differs from a standard news dashboard because it does not simply broadcast headlines; it frames every important development around four decision questions:

What changed? → Why does it matter? → Who should review it? → What action may be needed?

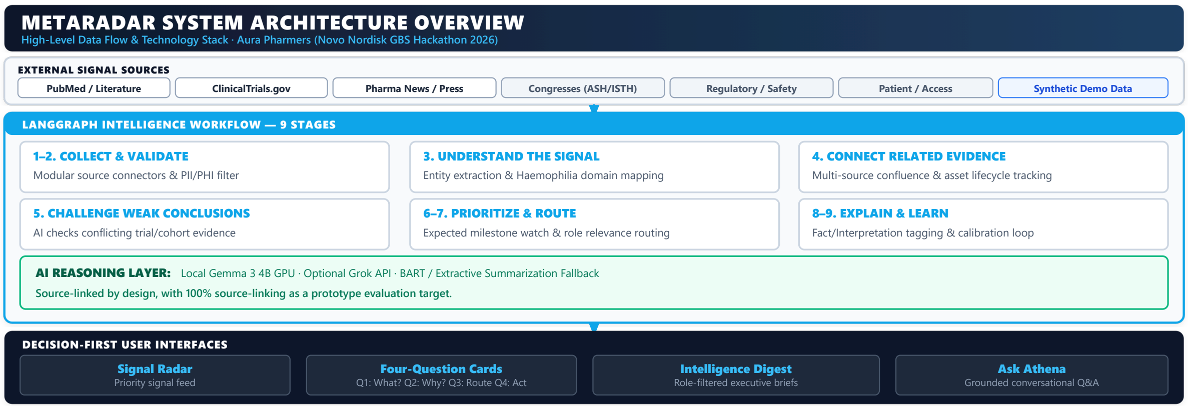
Intended users: Medical Affairs, Regulatory, Safety/Pharmacovigilance, Market Access, and Leadership. The system is decision support for expert review, not a replacement for professional judgment.

2. Overall concept, key features, architecture, and decision improvement

MetaRadar is an AI-enabled intelligence radar that ingests relevant external updates, connects related evidence, summarizes developments, assigns priority, and routes signals to the functions most likely to need review.

Detect	Identify relevant haemophilia developments across target channels.
Classify	Categorize clinical, publication, news, regulatory, safety, congress, and access signals.
Normalize	Identify therapies, companies, diseases, modalities, trial phase/status, and relevant entities.
Connect	Associate related evidence across sources to construct integrated signal trees.
Summarize	Generate concise, evidence-grounded explanations with source citations.
Prioritize	Assess relevance and urgency using explicit criteria and domain thresholds.
Route	Direct signals to the appropriate functional reviewers (Medical, Regulatory, Safety, Access).
Calibrate	Use structured stakeholder feedback to refine routing, scoring, and summary parameters.

Technology Stack & Architecture: Next.js 16 App Router (React 19, TailwindCSS) frontend; Python 3.11 FastAPI backend; PostgreSQL 16 with pgvector for relational/vector storage; Local Gemma 2 / Llama 3 for evidence-grounded RAG summarization and classification.



3. Haemophilia focus, data boundary, signal types, and out of scope

Focus	Haemophilia A and Haemophilia B within Rare Disease.
Signal types	Clinical-trial updates and milestones; scientific publications; company/pharma developments; regulatory decisions; safety/PV signals; congress developments; market-access and reimbursement updates.
Data boundary	Public, mock, or synthetic data only. No confidential Novo Nordisk information and no patient-identifiable private health information.
Out of scope	Patient-specific medical advice; autonomous clinical, medical, or regulatory decisions; replacement of expert review; confidential company data; autonomous external business actions; production enterprise deployment during the hackathon.

4. Public, mock, and synthetic data sources

Clinical trials	ClinicalTrials.gov for study identifiers, phase/status, milestones, endpoints, and study information.
Publications	PubMed / NCBI scientific literature.
Industry/news	NewsAPI plus controlled public/mock news feeds.
Regulatory/safety	OpenFDA / EMA RSS feeds plus mock or synthetic demonstration records.
Congress	Public or controlled data representing ASH, ISTH, WFH, EHA and similar meetings.
Access	Public/mock/synthetic reimbursement and access events.

Clinical-trial context: the referenced Novo Nordisk material describes clinical trials as protocol-driven studies used to investigate whether an investigational medication works, its safety, and side effects, with development progressing through Phases I–IV.

5. Detection, classification, summarization, priority, and traceability

Ingest → Detect → Classify → Deduplicate → Link Evidence → Summarize → Prioritize → Route → Human Review

Signals preserve stable source identifiers and provenance. Summaries remain linked to source evidence; the interface distinguishes source-supported facts from AI interpretation and forward-looking hypotheses. Sensitive medical, regulatory, and safety signals require human review.

6. Stakeholder input and AI calibration

Stakeholder input improves AI output by giving structured feedback on whether a signal was appropriately summarized, prioritized, and routed. The feedback informs later calibration without giving the AI autonomous decision authority.

AI Baseline → Expert Review → Structured Feedback → Calibrated Output

AI baseline	"A competing haemophilia therapy has generated significant clinical-trial activity."
Stakeholder-calibrated	**Priority: High.** Review by Medical Affairs and Competitive Intelligence; monitor subsequent efficacy, safety, and regulatory developments.

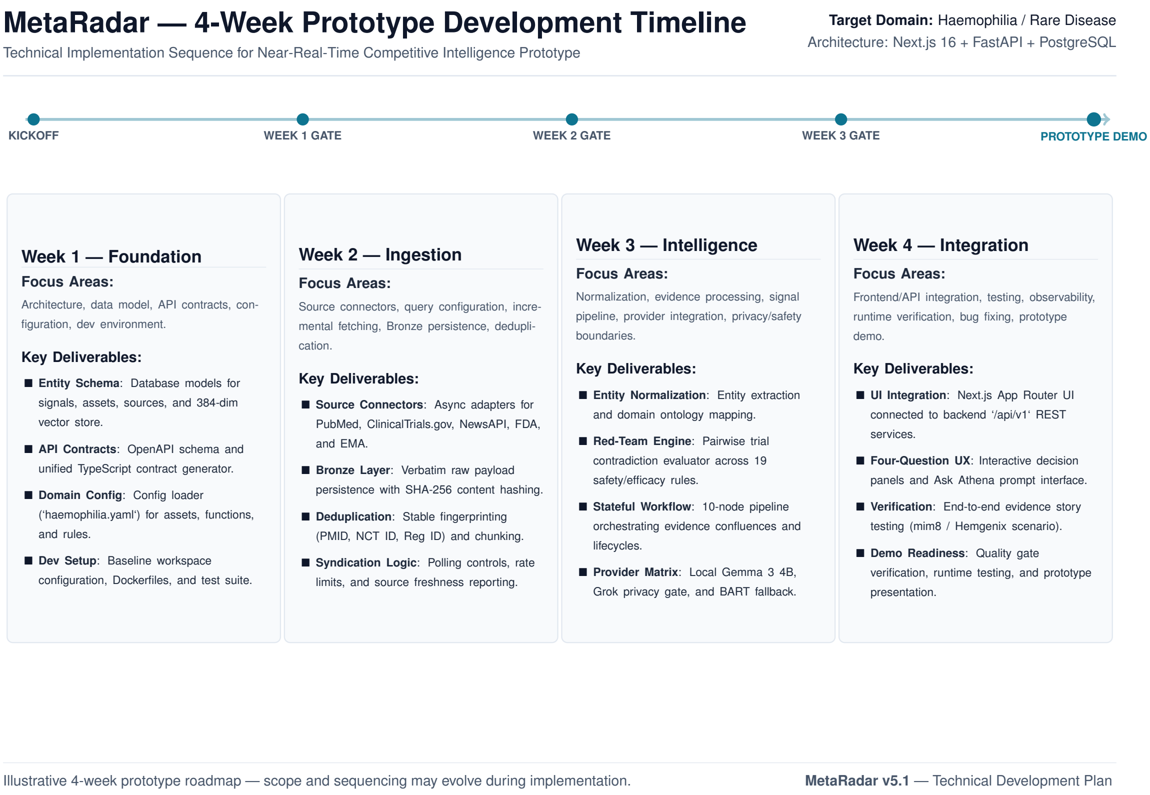
Radar Overview	High-priority developments, emerging clusters, and the current haemophilia landscape.
Signal Explorer	Search and filter by therapy, company, signal type, source, priority, date, and trial phase/status.
Signal Detail	"What changed?" "Why does it matter?" "Who should review?" "What action may be needed?" plus source evidence.
Evidence View	Supporting records, source links, timestamps, and provenance.
Calibration View	Stakeholder feedback on relevance, priority, routing, and summary quality.
Signal Card	Concise decision card combining summary, priority, reviewer, suggested next step, and evidence.

Role-based views

Medical Affairs	Clinical evidence, treatment landscape, efficacy, and safety developments.
Regulatory	Approvals, submissions, regulatory decisions, and regulatory/safety events.
Safety / PV	Safety signals, advisories, and emerging evidence requiring review.
Market Access	Access, reimbursement, affordability, and market-impact developments.
Leadership	High-priority developments, strategic context, and concise cross-source summaries.

7. Four-week prototype plan

Week	Focus	Deliverables & Milestone Scope
1	**Foundation & Setup**	Finalize taxonomy, DB schema (PostgreSQL 16), FastAPI core, Next.js UI shell, public/mock connectors.
2	**Ingestion & AI Pipeline**	Build connectors (ClinicalTrials, PubMed, NewsAPI, OpenFDA), detection, normalization, deduplication.
3	**Prototype Building**	Grounded summarization, priority scoring, routing logic, calibration loop, role views, signal cards.
4	**Testing & Demo Prep**	End-to-end integration, source-link provenance verification, calibration tuning, UI polish, final demo.



8. Clarifications and stakeholder input requested from Novo Nordisk

1. Which haemophilia signal categories should receive the highest baseline priority?
2. Which criteria should most influence urgency: trial phase/status, efficacy, safety, regulatory action, competitive relevance, access impact, or another factor?
3. How should prioritization and routing differ across Medical, Regulatory, Safety, Market Access, and Leadership?
4. What feedback format would be most useful and least intrusive for busy pharma professionals?
5. Which dashboard views and workflows would provide the greatest immediate decision-making value?

9. Expected outcome

MetaRadar reduces the gap between **information availability** and **decision relevance**. It does not replace experts or make decisions; it organizes relevant developments into a traceable workflow so the right function can review the right evidence at the right time.

Scattered information → Connected evidence → Prioritized intelligence → Human review → Better-informed action